

POKHARA UNIVERSITY

Level: Master

Semester: Spring

Year : 2026

Programme: ME/MSc Computer

Full Marks : 100

Course: Artificial Intelligence (New)

Pass Marks : 60

Time : 4 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1.

a) What is Simulated Annealing in local search? Explain its working mechanism and how it helps in escaping local minima. 7

b) Illustrate how the best first search may stuck in a loop. Also explain how this problem is resolved in A* search. 8
 2.

a) Consider the following knowledge base. Represent them in FOPL and use Resolution technique to reach goal: 8

 - i. Ram owns a dog.
 - ii. Cat is an animal.
 - iii. Every dog owner is animal lover.
 - iv. No animal lover can kill animals.
 - v. Either Ram or Sita killed a cat.

Goal: Who killed a cat?
 - b) Dynamic Bayesian Networks? Explain their structure and how they are used for probabilistic reasoning over time with a suitable example. 7
- OR**
- ✓ What is Ontological Engineering in AI? Explain its process and significance in knowledge representation with an example.
 3.

a) Explain STRIPS and PDDL in detail. Describe the components of a planning problem, including states, actions, preconditions, and effects, with a suitable example. 7

b) Explain Markov Decision Processes (MDPs) and the algorithms used to solve them. 8
 4.

a) Solve the following Map-Coloring Problem using Backtracking Search. 8



Variables WA, NT, Q, NSW, V, SA, T

Domains $D_i = \{red, green, blue\}$

Constraints: adjacent regions must have different colors

e.g., $WA \neq NT$ (if the language allows this), or

$(WA, NT) \in \{(red, green), (red, blue), (green, red), (green, blue), \dots\}$

- b) Why do you understand by Supervised, unsupervised and reinforcement learning? Discuss with appropriate applications. 7
5. a) Explain the architecture and mathematical formulation of a Graph Attention Network. 8

OR

Given input vector $x = [2, 3, -1]$, initial weights $= [0.4, -0.5, 0.2]$, bias $b = 0.6$, target output $y = 1$, and learning rate $\eta = 0.1$:

- Calculate the net input and output using a step activation function.
- Determine the classification error.
- Apply the Perceptron Learning Rule to update the weights and bias.
- Repeat the process for the input vector $[1, -2, 2]$ with target output 0.

- b) The Transformer architecture is a revolutionary deep learning framework that uses encoder decoder architecture. Are BERT and GPT related with it? Explain with block diagram. 7
6. a) Discuss about the different type of word embedding strategies along with examples. 7
- b) You have been assigned to build a grievance handling chatbot using large language models (LLMs). How will you develop the system and what will be the role of Pretraining and Finetuning in this scenario. Explain. 8

7. Write short notes on: (Any two)

- a) CNN
- b) Transfer Learning
- c) Challenges in NLP